



“Seminars in Biomedical Engineering”

Programa de Pós-Graduação em Engenharia Biomédica

11/10/2021 – 13h00

Google Meet Link: <http://meet.google.com/ukv-ieyy-bfe>

THE AUDITORY APPARATUS: FROM PHYSIOLOGY TO MICRO-PROSTHETICS

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Abstract

Hearing is the sense devoted to perceiving sounds and, therefore, is an important mean for social inclusion. Although a complete understanding of the physiological mechanisms of the auditory apparatus is still missing, the World Health Organization has reported an increasing demand for prostheses able to restore the hearing sense. This seminar will provide an overview of the fundamental physiological mechanisms of the ear and deliver some of most recent studies on bioengineering approaches for ear prosthetics, giving a synergistic overview of the interdisciplinary aspects involved in the field, from physiology to materials science and micro-fabrication. Finally, specific attention will be given to modeling, one of the most relevant and effective tools to deal with such a complex and scientific interconnected field.

Speaker's Bio

Dr. Mario Milazzo is an Assistant Professor at University of Pisa. He graduated in 2010 in Mechanical Engineering at University of Pisa (Pisa, Italy) and received his PhD in BioRobotics in 2016 at the BioRobotics Institute of Scuola Superiore Sant'Anna (Pisa, Italy) working in the micro-prosthetics field. His

research interests cover a large number of fields: multiscale modeling, mechanics of acoustics, biomaterials, mechatronics, and industrial robotics. He has been working, as Visiting Scientist, at Columbia University (New York, NY USA) and Massachusetts Institute of Technology (Cambridge, MA, USA). He currently holds two further appointments with Scuola Superiore Sant'Anna and Massachusetts Institute of Technology. He has been the recipient of the 3-year Marie Skłodowska-Curie Individual Fellowships (MSCA-IF-GF) in collaboration with Massachusetts Institute of Technology (USA), University of Antwerp (Belgium), Tufts University (USA), Massachusetts Eye&Ear (USA) and University of Pisa (Italy). He has co-authored 40+ papers on international journals, 5+ book chapters, 9 patents, and 30+ contributions to national/international conferences and workshops, serving also as an invited and keynote speaker. In parallel to the academic activity, he has been involved, both as a Project Manager and Team Leader, in several projects and collaborations with national and international companies (e.g., Baker Hughes, Brembo, Magneti Marelli, Piaggio).